

Software Requirements Specification

IICT LIBRARY MANAGEMENT SYSTEM

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1.Introduction

The IICT Library Management System is a web-based software solution developed to support and modernize the daily operational activities of the IICT Library. Traditionally, the IICT Library relies heavily on manual processes such as handwritten registers, logbooks, and physical borrower lists to manage books, students, and faculty borrowing records. These manual approaches are time-consuming, error-prone, and difficult to maintain over long periods.

This system aims to digitize those processes while preserving the real-life workflow of the IICT Library. It provides a centralized platform where all library-related data—books, borrowers, transactions, and monitoring records—can be securely stored, accessed, and managed. The system is designed to serve only authorized members of IICT, ensuring proper control and accountability of library resources.

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to clearly define and describe the functional and non-functional requirements of the IICT Library Management System. This document serves as a formal reference for developers, project supervisors, testers, and academic evaluators. It translates the business-level requirements outlined in the BRS into detailed system requirements that guide system design, development, and validation.

The SRS ensures that all stakeholders share a common understanding of what the system will do, how it will behave, and what constraints it must operate under. It also acts as a baseline document against which the final system can be evaluated.

1.2 Scope

The scope of the IICT Library Management System includes the complete digitization of library book management and borrowing processes within the IICT departments. The system supports two types of users: **Admin (Library Authority)** and **Student/Teacher (Borrowers)**.

The system manages detailed book information, including accession numbers, bibliographic details, and physical classification using the Dewey Decimal Classification (DDC) system. It also manages student and faculty borrowing records, tracks issued and returned books, and monitors outside books brought into the library for in-house reading.

The system is strictly limited to academic use within IICT and is applicable to the CSE, SWE, and EEE departments only.

1.2.1 Product Overview

The IICT Library Management System is a centralized, browser-based application that mirrors the real operational practices of the IICT Library in a digital form. It provides a secure administrative interface for managing library assets and a user-friendly interface for students and teachers to access library services.

The system maintains accurate records of books from acquisition to circulation and return. By replacing physical registers with digital records, the system improves efficiency, reduces redundancy, and enhances data reliability while maintaining transparency and accountability.

1.2.2 Product Features

The system provides role-based access and functionality tailored to different user types. Core features include secure authentication, comprehensive book catalog management, borrowing and return tracking, and record maintenance for both students and faculty.

The system supports detailed book metadata such as title, author, edition, volume, ISBN, publisher, publication date, binding type, pagination, source of acquisition, bill information, accession number, and DDC classification. Borrowing features include issuing books through accession numbers, updating availability status, and maintaining borrowing histories.

Additional features include a dedicated outside-book monitoring mechanism for students, separate borrower records for faculty members, and administrative reporting tools to monitor library usage.

1.3 Definitions, Acronyms, and Abbreviations

SRS: Software Requirements Specification

BRS: Business Requirements Specificatio

IICT: Institute of Information and Communication Technology

Admin: Authorized library authority responsible for managing the system

Student/Teacher: IICT students or faculty members eligible to borrow books

Accession Number: A unique identification number assigned to each book

DDC: Dewey Decimal Classification system used for subject-based organization

ISBN: International Standard Book Number used to uniquely identify books

LMS: Library Management System.

1.4 References

IEEE Std 29148-2018 – Software Requirements Specification

IICT Library Management System – Business Requirements Specification

Ian Sommerville, *Software Engineering*

1.5 Overall Description

The IICT Library Management System functions as a reliable and secure platform for managing library operations. It enables administrators to control book inventories, track borrowing activities, and maintain accurate records. Students and teachers interact with the system primarily to search for books and borrow or return them following library policies.

The system ensures data consistency, enforces access control, and reflects real-life verification processes such as ID-based borrowing and faculty authorization. By digitizing these processes, the system reduces manual effort while preserving operational integrity.

2.1 Product Perspective

From a system perspective, the IICT Library Management System is a standalone web application connected to a centralized database. All user interactions occur through a web interface, while data storage and processing are handled on the server side.

The system replaces traditional paper-based registers such as book entry logs, borrower lists, faculty registers, and outside-book monitoring logs. It integrates these functions into a unified digital system without requiring integration with external third-party systems.

2.2 User Characteristics

Admin users are library staff members responsible for managing books, verifying borrowers, maintaining records, and overseeing overall library operations. They are expected to have basic computer literacy and familiarity with library rules and procedures.

Student and Teacher users are authorized members of IICT who borrow books for academic purposes. They require a simple, intuitive interface to search books, view availability, and manage their borrowing history. Teachers typically borrow books for extended academic use, while students may also register outside books for reading within library premises.

2.3 Constraints

The system operates under several constraints, including

Role-Based Access Control: Users can only perform actions permitted by their roles (e.g., Admin, Student, Teacher).

Internet Connectivity Dependence: The system requires a stable internet connection for all operations.

Compliance with Policies: The system must adhere to IICT academic regulations and library policies.

Data Security: The system must prevent unauthorized access and ensure the confidentiality and integrity of user and library data.

Web-Based Operation: The system is designed to function as a web application and cannot operate offline.

Server Availability: Proper functioning depends on continuous availability of the server infrastructure.

Database Dependency: All operations rely on database availability; any downtime may prevent borrowing, returning, or searching books

2.4 Assumptions and Dependencies

The system operates under the following assumptions and dependencies:

Valid User Credentials: All users (students, teachers, and staff) are assumed to have valid IICT identification credentials

Internet Access: Users are assumed to have reliable internet connectivity to access the web-based system.

Staff Training: Library staff and administrators are assumed to be trained to use the system's administrative functions effectively.

Web Hosting Availability: The system depends on continuous availability of the web hosting infrastructure.

Database Services: The system relies on database services for storing and retrieving data; database downtime may impact functionality.

System Maintenance: Proper and regular maintenance of server, software, and hardware components is assumed for optimal system performance.

User Compliance: Users are assumed to comply with IICT policies and follow system guidelines while using the library system.

3. Specific Requirements

This section specifies the functional and non-functional requirements and user interface requirements of the IICT Library Management System in a structured and detailed format, following the same style as the provided sample SRS.

3.1 Functional Requirements

ID: LMS-FR1

Title: User Authentication and Login

Description: The system must provide secure authentication facilities for Admin and Student/Teacher users.

Rationale: To ensure that only authorized IICT members can access the library system.

Dependencies: User credential database

Scenario: A user enters valid credentials and successfully logs into the system.

Scenario Type: Security Feature

Scenario Description: An admin logs into the system and gains access to the administrative dashboard.

ID: LMS-FR2

Title: Role-Based Access Control

Description: The system must restrict features based on user roles.

Rationale: To prevent unauthorized access to administrative functions.

Dependencies: Authentication module

Scenario: A student attempts to access admin features and is denied.

Scenario Type: Security Feature

Scenario Description: A student logs in and can only access book search and borrowing features.

ID: LMS-FR3

Title: Book Record Creation

Description: The system must allow Admin users to add new book records.

Rationale: To maintain an up-to-date digital book catalog.

Dependencies: Book database

Scenario: An admin enters new book details and saves them.

Scenario Type: Data Management

Scenario Description: A librarian adds a newly purchased textbook into the system.

ID: LMS-FR4

Title: Book Information Update

Description: The system must allow Admin users to update existing book records.

Rationale: To correct or modify book information when necessary.

Dependencies: Book database

Scenario: An admin edits book details and saves changes.

Scenario Type: Data Management

Scenario Description: A librarian updates the edition number of a book.

ID: LMS-FR5

Title: Book Deletion

Description: The system must allow Admin users to remove book records.

Rationale: To remove damaged or obsolete books from the system.

Dependencies: Book database

Scenario: An admin deletes a book record.

Scenario Type: Data Management

Scenario Description: A librarian removes a damaged book from inventory.

ID: LMS-FR6

Title: Book Classification Using DDC

Description: The system must support Dewey Decimal Classification for books.

Rationale: To reflect the physical organization of the IICT library.

Dependencies: Classification module

Scenario: An admin assigns a DDC number to a book.

Scenario Type: Classification Feature

Scenario Description: A librarian assigns a DDC code while cataloging a book.

ID: LMS-FR7

Title: Book Search and Viewing

Description: The system must allow users to search and view book details.

Rationale: To help users find required books efficiently.

Dependencies: Book database

Scenario: A user searches by title or author.

Scenario Type: User Interaction

Scenario Description: A student searches for a programming book and views availability.

ID: LMS-FR8**Title:** Book Issuing Process**Description:** The system must allow Admin users to issue books to borrowers.**Rationale:** To digitally record book borrowing.**Dependencies:** Borrower and book database**Scenario:** An admin issues a book using accession number.**Scenario Type:** Transaction Processing**Scenario Description:** A librarian issues a book to a teacher.**ID:** LMS-FR9**Title:** Book Return Processing**Description:** The system must support book return operations.**Rationale:** To maintain accurate availability records.**Dependencies:** Borrowing records**Scenario:** A book is returned and marked available.**Scenario Type:** Transaction Processing**Scenario Description:** A student returns a borrowed book.**ID:** LMS-FR10**Title:** Student Borrower Record Management**Description:** The system must maintain borrowing records for students.**Rationale:** To track student borrowing history.**Dependencies:** Student database**Scenario:** A student borrows multiple books.**Scenario Type:** Record Management**Scenario Description:** The system updates a student's borrowing history.**ID:** LMS-FR11**Title:** Faculty Borrower Record Management

Description: The system must maintain separate borrowing records for teachers.

Rationale: To distinguish faculty borrowing privileges.

Dependencies: Faculty database

Scenario: A teacher borrows books.

Scenario Type: Record Management

Scenario Description: A faculty member's borrowing record is updated.

ID: LMS-FR12

Title: Outside Book Entry

Description: The system must allow students to register outside books.

Rationale: To monitor non-library books brought into the library.

Dependencies: Monitoring module

Scenario: A student logs an outside book.

Scenario Type: Monitoring Feature

Scenario Description: A student registers an external book for reading.

ID: LMS-FR13

Title: Outside Book Verification

Description: The system must allow Admin to verify outside book entries.

Rationale: To prevent unauthorized book removal.

Dependencies: Monitoring records

Scenario: An admin verifies outside book exit.

Scenario Type: Security Feature

Scenario Description: A librarian verifies a student's outside book.

ID: LMS-FR14

Title: Borrowing History View

Description: The system must display borrowing history to users.

Rationale: To provide transparency to borrowers.

Dependencies: Borrowing database

Scenario: A user views past borrowed books.

Scenario Type: User Interaction

Scenario Description: A student reviews previously borrowed books.

ID: LMS-FR15

Title: Report Generation

Description: The system must generate administrative reports.

Rationale: To assist library management decisions.

Dependencies: Reporting module

Scenario: An admin generates an issued-book report.

Scenario Type: Reporting Feature

Scenario Description: A librarian reviews daily issued book reports.

3.2 Non-Functional Requirements

ID: LMS-NFR1

Title: System Security

Description: The system must ensure secure access through encrypted authentication mechanisms.

Rationale: To protect sensitive user and library data from unauthorized access.

Dependencies: Authentication and encryption modules

Scenario: A user attempts to log in with valid credentials.

Scenario Type: Security Requirement

Scenario Description: The system validates credentials securely and grants access.

ID: LMS-NFR2

Title: Data Privacy

Description: The system must protect personal and borrowing data of users.

Rationale: To ensure confidentiality of student and teacher information.

Dependencies: Database security policies

Scenario: User data is stored and retrieved securely.

Scenario Type: Privacy Requirement

Scenario Description: Borrowing records are accessible only to authorized users.

ID: LMS-NFR3

Title: Performance Efficiency

Description: The system must respond to user requests within acceptable time limits.

Rationale: To ensure smooth and uninterrupted user experience.

Dependencies: Server performance and optimization

Scenario: A user searches for a book.

Scenario Type: Performance Requirement

Scenario Description: Search results are displayed within two seconds.

ID: LMS-NFR4

Title: System Availability

Description: The system must be available during official library working hours.

Rationale: To support uninterrupted academic access.

Dependencies: Hosting and server infrastructure

Scenario: Users access the system daily.

Scenario Type: Availability Requirement

Scenario Description: The system remains operational throughout the day.

ID: LMS-NFR5

Title: Reliability

Description: The system must operate consistently without unexpected failures.

Rationale: To maintain trust in digital library operations.

Dependencies: Stable software architecture

Scenario: Continuous system usage.

Scenario Type: Reliability Requirement

Scenario Description: Borrowing records are stored correctly without data loss.

ID: LMS-NFR6

Title: Usability

Description: The system must provide an intuitive and easy-to-use interface.

Rationale: To reduce training requirements for users.

Dependencies: UI design standards

Scenario: A first-time user navigates the system.

Scenario Type: Usability Requirement

Scenario Description: A student searches and borrows a book without assistance.

ID: LMS-NFR7

Title: Accessibility

Description: The system must be accessible from standard web browsers.

Rationale: To allow wide accessibility for all users.

Dependencies: Browser compatibility

Scenario: Users access the system from different devices.

Scenario Type: Accessibility Requirement

Scenario Description: The system functions correctly on common browsers.

ID: LMS-NFR8

Title: Maintainability

Description: The system must be easy to maintain and modify.

Rationale: To support future updates and bug fixes.

Dependencies: Modular codebase

Scenario: Developers update system features.

Scenario Type: Maintenance Requirement

Scenario Description: New features are added without affecting existing ones.

ID: LMS-NFR9

Title: Scalability

Description: The system must support growth in users and book records.

Rationale: To accommodate future expansion.

Dependencies: Database design

Scenario: Increased number of users.

Scenario Type: Scalability Requirement

Scenario Description: The system handles additional data smoothly.

ID: LMS-NFR10

Title: Data Backup and Recovery

Description: The system must support regular data backup and recovery.

Rationale: To prevent data loss.

Dependencies: Backup mechanisms

Scenario: System failure occurs.

Scenario Type: Recovery Requirement

Scenario Description: Data is restored from backup.

ID: LMS-NFR11

Title: Error Handling

Description: The system must handle errors gracefully.

Rationale: To avoid system crashes and confusion.

Dependencies: Exception handling mechanisms

Scenario: Invalid input is provided.

Scenario Type: Error Handling Requirement

Scenario Description: The system displays a clear error message.

ID: LMS-NFR12

Title: Audit and Logging

Description: The system must log important user and system activities.

Rationale: To support monitoring and accountability.

Dependencies: Logging module

Scenario: An admin performs system actions.

Scenario Type: Auditing Requirement

Scenario Description: Actions are recorded in system logs.

ID: LMS-NFR13

Title: Compatibility

Description: The system must be compatible with common operating systems.

Rationale: To ensure wide usability.

Dependencies: Web technologies

Scenario: Users access the system from different OS.

Scenario Type: Compatibility Requirement

Scenario Description: The system functions properly across platforms.

ID: LMS-NFR14

Title: Data Integrity

Description: The system must ensure accuracy and consistency of stored data.

Rationale: To maintain reliable records.

Dependencies: Database constraints

Scenario: Book records are updated.

Scenario Type: Data Integrity Requirement

Scenario Description: No conflicting or duplicate data is stored.

ID: LMS-NFR15

Title: Compliance

Description: The system must comply with IICT academic and library policies.

Rationale: To ensure institutional alignment.

Dependencies: Institutional guidelines

Scenario: System operation follows library rules.

Scenario Type: Compliance Requirement

Scenario Description: Borrowing limits are enforced correctly.

3.3 User Interface Requirements

The IICT Library Management System requires several user interfaces to support the complete workflow of library operations. Each interface is designed to be simple, consistent, and role-specific so that users can perform their tasks efficiently with minimal training.

Login Interface

The login interface is the entry point of the system for all users. It allows Admins and Student/Teacher users to enter their credentials and access the system based on their roles. The interface contains input fields for username or ID and password, along with a login button and error message display area. Upon successful authentication, users are redirected to their respective dashboards. This interface ensures secure access and prevents unauthorized users from entering the system.

The image shows a web form titled "IICT Library Login". It has a light beige background. The form contains two input fields: "Email / ID" with the text "redwan98" and "Password" with masked characters "*****". Below these is a dark brown "Login" button. At the bottom, there is a link "Don't have an account? [Register here](#)" and a "← Back to Home" link.

Fig: Login interface

Registration Interface:

The registration interface allows new Student and Teacher users to create an account in the system. It provides input fields for entering required information such as name, ID, department, email, and password, along with a registration submission button. The interface validates user input to ensure accuracy and completeness before account creation. Upon successful registration, user information is securely stored in the system database, and the user is redirected to the login interface. This interface ensures that only authorized IICT members can register and access the system.



The registration form is titled "IICT Library Registration" and contains the following fields:

- Full Name ***: A text input field with the placeholder "Enter your full name".
- Username ***: A text input field containing the value "redwan98".
- Email ***: A text input field with the placeholder "your.email@example.com".
- Phone Number**: A text input field with the placeholder "01XXXXXXXXXX".
- Registration Number ***: A text input field with the placeholder "e.g., 2020131047".

Fig: Registration interface

Admin Dashboard Interface

The Admin dashboard provides an overview of library activities after successful login. It displays key information such as total number of books, currently issued books, available books, student and faculty borrowing statistics, and alerts related to issued or returned books. From this dashboard, the Admin can navigate to all administrative functions including book management, borrower records, issuing and returning books, outside book monitoring, and report generation. The dashboard acts as the central control panel for library authorities.

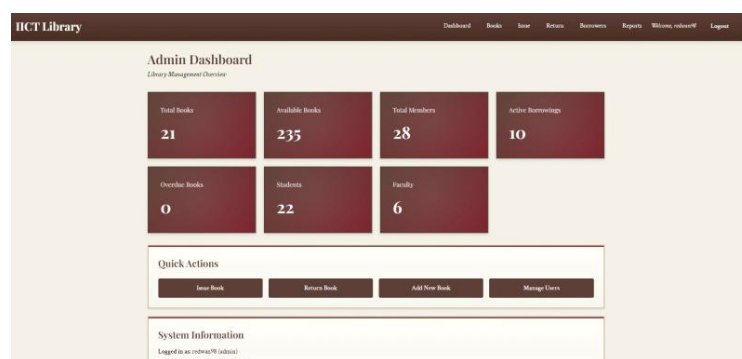


Fig: Admin Dashboard interface

Student Dashboard Interface

The student dashboard interface serves as the main workspace for students after successful login. It displays student-specific information such as profile details, library activities, issued books, and outside book registration status. The dashboard provides easy navigation to key functionalities, including registering outside books, viewing entry and exit records, and updating profile information. This interface is designed to be user-friendly and informative, enabling students to efficiently interact with the library system and perform their permitted actions.

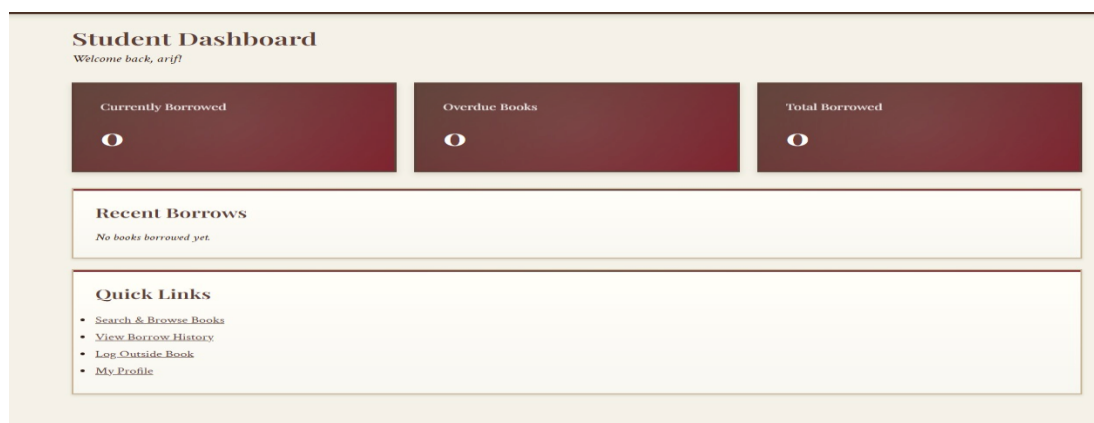
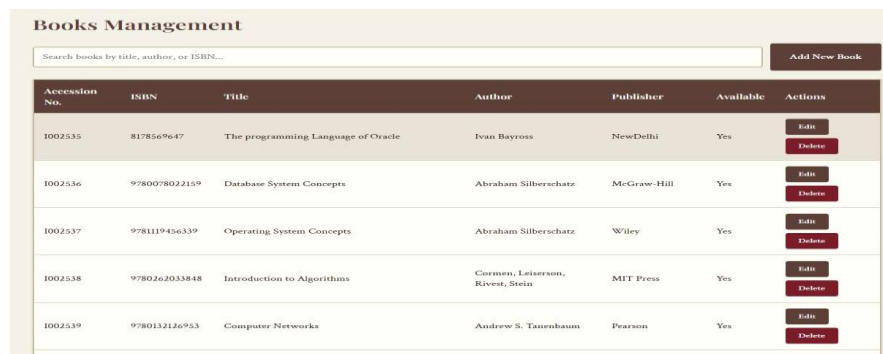


Fig: Student Dashboard interface

Book Management Interface

The book management interface is used by Admin users to maintain the library book catalog. It includes forms for adding new books and updating or deleting existing book records. The interface allows the Admin to enter detailed book information such as title, author, edition, publisher, ISBN, accession number, binding type, pagination, source of acquisition, bill details, and Dewey Decimal Classification number. This interface ensures accurate and organized digital cataloging of library resources.

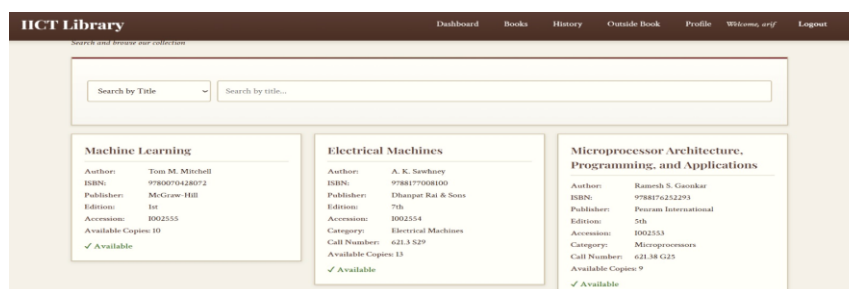


Accession No.	ISBN	Title	Author	Publisher	Available	Actions
1002535	8178569647	The programming Language of Oracle	Ivan Bayross	NewDelhi	Yes	Edit Delete
1002536	9780078022159	Database System Concepts	Abraham Silberschatz	McGraw-Hill	Yes	Edit Delete
1002537	9781119456339	Operating System Concepts	Abraham Silberschatz	Wiley	Yes	Edit Delete
1002538	9780262033848	Introduction to Algorithms	Cormen, Leiserson, Rivest, Stein	MIT Press	Yes	Edit Delete
1002539	9780132126953	Computer Networks	Andrew S. Tanenbaum	Pearson	Yes	Edit Delete

Fig: Book management interface

Book Search and Listing Interface

The book search interface is accessible to both Admin and Student/Teacher users. It allows users to search for books using different criteria such as title, author, subject, accession number, or classification number. Search results are displayed in a structured list showing basic book details and availability status. Users can select a book from the list to view detailed information. This interface helps users quickly locate required books within the library.



Search by Title	Search by title...
Machine Learning Author: Tom M. Mitchell ISBN: 9780070428072 Publisher: McGraw-Hill Edition: 1st Accession: 1002535 Available Copies: 10 ✓ Available	Electrical Machines Author: A. K. Sawhney ISBN: 9788177008100 Publisher: Thompson Rat & Sons Edition: 7th Accession: 1002554 Category: Electrical Machines Call Number: 621.3 S29 Available Copies: 13 ✓ Available
Microprocessor Architecture, Programming, and Applications Author: Ramesh S. Gaonkar ISBN: 9788176232293 Publisher: Pearson International Edition: 5th Accession: 1002533 Category: Microprocessors Call Number: 621.38 G25 Available Copies: 9 ✓ Available	

Fig: Book search and listing interface

Book Details Interface

The book details interface displays complete information about a selected book. It shows bibliographic details, physical description, accession number, DDC classification, and current availability status. For Admin users, this interface also provides options to issue, update, or remove the book. For Student/Teacher users, it allows them to check whether the book is available for borrowing. This interface supports informed decision-making before borrowing.

← Back to Catalog

Bibliographic Information	Availability
Title:	Total Copies:
Author:	Available:
ISBN:	Staff Location: N/A
Publisher: N/A	X Currently Unavailable
Publication Year: N/A	
Category: N/A	

Note: To borrow this book, please visit the library counter with your student ID card.
Borrowing Period: 14 days | Fine for late return: €5 per day

Fig: Book Details interface

Book Issuing Interface

The book issuing interface is used exclusively by Admin users to issue books to students or teachers. The interface allows the Admin to select a borrower, enter or scan the accession number of the book, and confirm the issuing transaction. Once issued, the system updates the book's availability status and records the transaction in the borrower's history. This interface ensures controlled and verified book circulation.

Issue Book

Select Book *

Choose a book...

Search Member (Student/Faculty) *

Search by name, reg number, or teacher ID

Borrow Date *

01/10/2026

Issue Book

Fig: Book Issue interface

Book Return Interface

The book return interface allows Admin users to process returned books. The Admin enters the accession number of the returned book, and the system updates the book's status to available. The borrower's record is also updated automatically. This interface ensures that returned books are correctly tracked and made available for future borrowing.



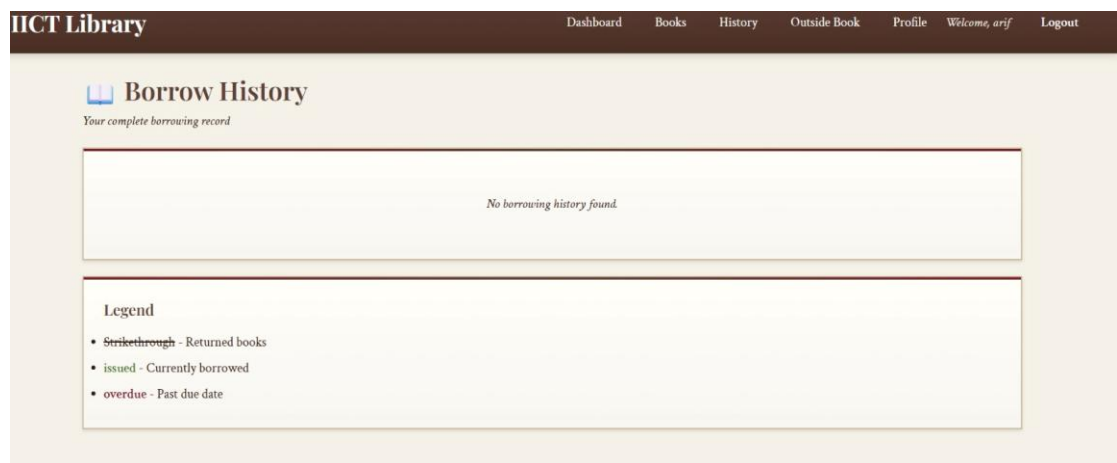
The screenshot shows a web interface titled "Return Book". At the top, there is a search bar with the placeholder text "Search by member name, ID, or book title...". Below the search bar is a table with five columns: Member, Book, Borrow Date, Status, and Action. The table contains seven rows of data, each representing a book borrowing record. The "Status" column uses color-coded labels: "Borrowed" in yellow and "Issued" in light blue. The "Action" column contains a red button labeled "Mark Returned" for each row.

Member	Book	Borrow Date	Status	Action
Md Redwan Hasan 2020131047 (student)	The programming Language of Oracle 1002535	1/10/2026	Borrowed	Mark Returned
Imran Hossain 2020631052 (student)	Artificial Intelligence: A Modern Approach 1002545	8/18/2025	Issued	Mark Returned
Sadia Akter 2020431050 (student)	Signals and Systems 1002543	8/14/2025	Issued	Mark Returned
Md Redwan Hasan 2020131047 (student)	Digital Design 1002541	8/10/2025	Issued	Mark Returned
Rumana Akter 2018811048 (student)	Computer Networks 1002539	8/7/2025	Issued	Mark Returned
Mahfuz Ahmed 2018431051 (student)	Operating System Concepts 1002537	8/3/2025	Issued	Mark Returned
Dr Ifte Khairul Amin eee5585 (faculty)	Principles of Electrical Engineering 1002547	8/11/2025	Borrowed	Mark Returned

Fig: Book return interface

Student/Faculty Borrowing History Interface

This interface allows students to view their borrowing history. It displays a list of books currently issued and previously returned, along with issue and return dates. This interface provides transparency to students regarding their borrowing activities and helps them track borrowed resource



The screenshot shows a web interface titled "Borrow History" under the "HICT Library" header. The header includes navigation links: Dashboard, Books, History, Outside Book, Profile, Welcome, arif, and Logout. The main content area has a sub-header "Borrow History" with the tagline "Your complete borrowing record". Below this is a large empty box with the text "No borrowing history found." at the bottom. At the bottom of the interface is a "Legend" section with three items: "Strikethrough - Returned books", "issued - Currently borrowed", and "overdue - Past due date".

Fig: Borrow History interface

Outside Book Entry Interface

The outside book entry interface allows students to register books that are brought from outside the library for in-library reading. Students enter basic details of the outside book and submit the entry before using the book inside the library. This interface supports monitoring and prevents unauthorized removal of library books.

Outside Book Entry
Log books you're bringing from outside the library

Note: Students bringing books from outside must register them here for security and tracking purposes. This helps maintain library records and prevents confusion with library books.

Registration Number
Enter your registration number

Book Title
Enter the book title

Department
Select Department

Semester
Select Semester

Log Outside Book

Fig: Outside Book entry interface

4. Prioritization and Release Plan

4.1 Choice of prioritization method

The prioritization of features and requirements for the IICT Library Management System (LMS) is essential for ensuring that the most critical functionalities are delivered first, while also considering dependencies and technical feasibility. We have used the following approach for prioritization:

MoSCoW Method:

- **Must Have:** These are the core features that are absolutely necessary for the system to function effectively. These include:
 -
 - **Member Registration and Management:** Handling CSE, SWE, and EEE student and faculty records.
 - **Inventory & Catalog Management:** Maintaining book records with unique Accession Numbers and Spine labels.
 - **Circulation Management:** The core borrowing and returning logic (including the "striketrough" completion method).
- **Should Have:** These are important features that enhance system functionality but are not critical for the initial release. These include:
 - **Outside Book Entry Tracking:** Monitoring personal books brought into the library by students.
 - **Reporting Module:** Generating basic audit reports and usage statistics for administration.
- **Could Have:** These features are nice-to-have and can be included in future updates if resources allow. These include:
 - **Automated Email/SMS Notifications:** For overdue book reminders.

- **Digital Spine Label Generator:** A tool to automatically format and print labels for new books.
- **Won't Have:** These features are not necessary for the initial release but could be considered in future versions. Examples include:
 - **E-book Reader Integration:** Support for reading digital copies within the system.
 - **AI-based Book Recommendations:** Suggesting books based on a student's borrowing history.

Priority Chart for IICT Library Management System (LMS):

Priority	Feature	Description
Must Have	Member Management	Register and manage Students/Faculty with Dept and ID details.
	Catalog Management	Store book info (Title, Author, Accession No, ISBN) and Spine info.
	Circulation System	Handle borrowing/returning using Accession No tracking.
	Search Functionality	Allow users to search for books by title, author, or category.
Should Have	Outside Book Log	Digital log for students to record personal books brought inside.
	Overdue Tracking	Identify books not returned by the due date.
	Basic Reporting	Generate reports on total books, active loans, and member lists.
Could Have	Notification System	Automated alerts for overdue books or reservation arrivals.
	Fine Management	Calculation of fines for late returns based on library policy.
Won't Have	Advanced Analytics	Predict future book demand or student study patterns using AI.
	Integrated E-Library	Full-text digital access to all physical books in the library.

4.2 Release Plan

The IICT LMS will be released in multiple phases, ensuring that essential functionalities are available early on while allowing for iterative improvements and feature additions.

Release 1: Core Operations

- **Features:**
 - Member Registration (Students and Faculty).

- Book Inventory/Catalog Setup.
 - Borrowing and Returning Module.
 - Basic Book Search.
- **Objective:** Deliver the essential features required to replace manual logs for book transactions and member records.

Release 2: Security and Management

- **Features:**
 - Outside Book Entry Tracking system.
 - Overdue book tracking and status updates.
 - Spine label management for physical organization.
- **Objective:** Enhance library security and organization by tracking personal items and ensuring books are returned on time.

Release 3: Reporting and Refinement

- **Features:**
 - Reporting Module for Administration (Audit and Usage).
 - Continuous Monitoring and Bug Fixes.
 - Integration of automated notifications/alerts.
- **Objective:** Address user feedback from the first two releases and provide administrative tools for long-term system oversight.

5. System Models

5.1 Use Case Diagrams:

5.1.1 Level-0 Use Case Diagram (Overall System)

Description

This level of use case diagram describes the overall IICT Library Management System and the actors interacting with it. The system is accessed by two primary actors who perform different operations based on their roles.

Actors

- Admin (Library Authority)
- Student
- Teacher

Tasks

- Authenticate User
- Manage Book Records

- Borrow and Return Books
- Search and View Books
- Manage Outside Book Records
- View Borrowing History

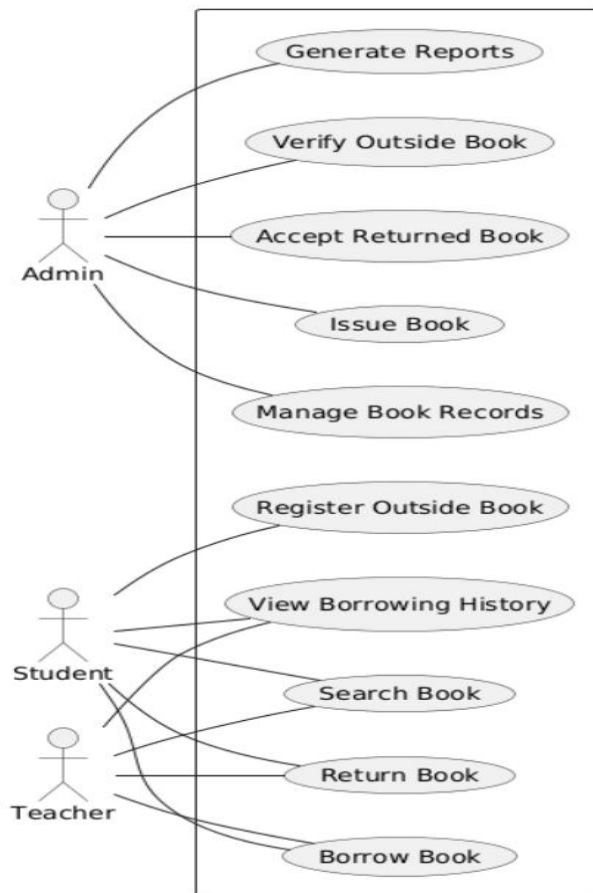


Fig: Overall system use case Diagram(level 0)

5.1.2 Use Case Diagram and Description of Subsystems

This section describes the major subsystems of the IICT Library Management System and the role of each subsystem in the overall system.

5.1.2.1 Book Entry and Management Subsystem (Admin)

Description

This subsystem manages all book-related administrative operations including adding, updating, classifying, and removing books from the library database.

Tasks

- Add New Book
- Update Book Information

- Delete Book
- Assign Accession Number
- Assign DDC Classification

Actors

- Admin

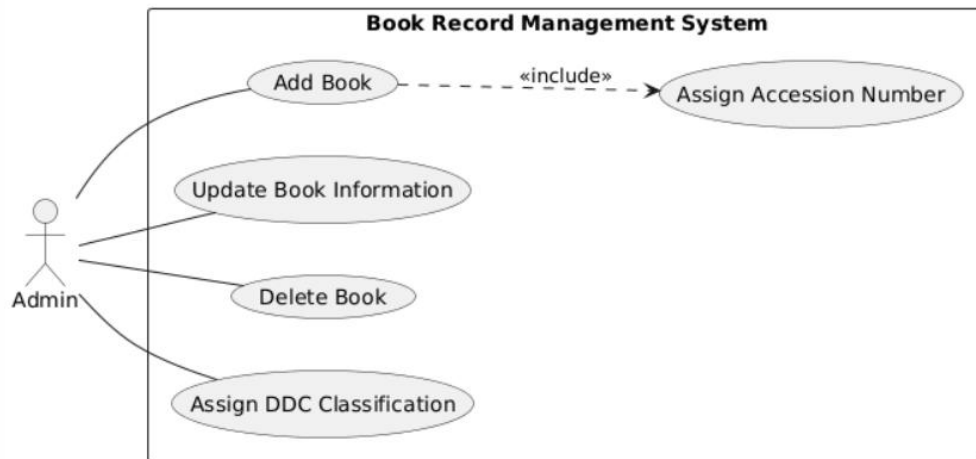


Fig: Book entry/management use case diagram

5.1.2.2 Borrowed Books Management (Students & Teachers)

Description

This subsystem handles the borrowing and returning of books by students and teachers. It maintains borrowing records and updates book availability.

Tasks

- Search Book
- Borrow Book
- Return Book
- View Borrowing History

Actors

- Student
- Teacher
- Admin

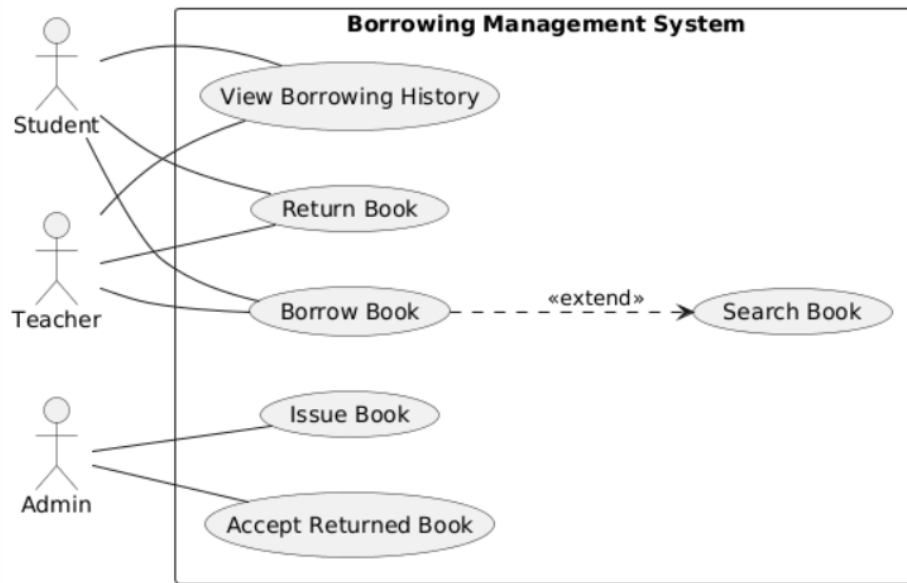


Fig: Book borrowing use case diagram

5.1.2.3 Outside Book Entry Subsystem

Description: This subsystem monitors books brought from outside the library by students for in-library reading and ensures security verification during exit.

Tasks:

- Register Outside Book
- Verify Outside Book
- Maintain Outside Book Log

Actors

- Student
- Admin

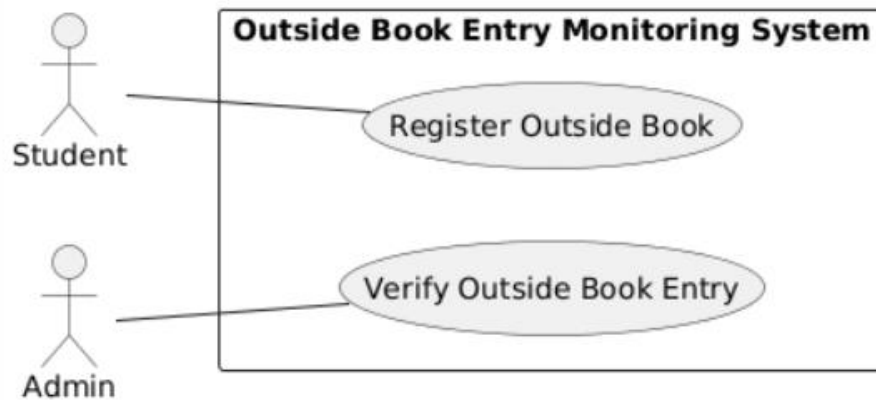


Fig: Outside Book entry use case diagram

5.1.2.4 Book Search and Viewing Subsystem

Description

This subsystem allows users to search, view, and check availability of library books using different search criteria.

Tasks

- Search Book
- View Book Details
- Check Availability

Actors

- Admin
- Student
- Teacher

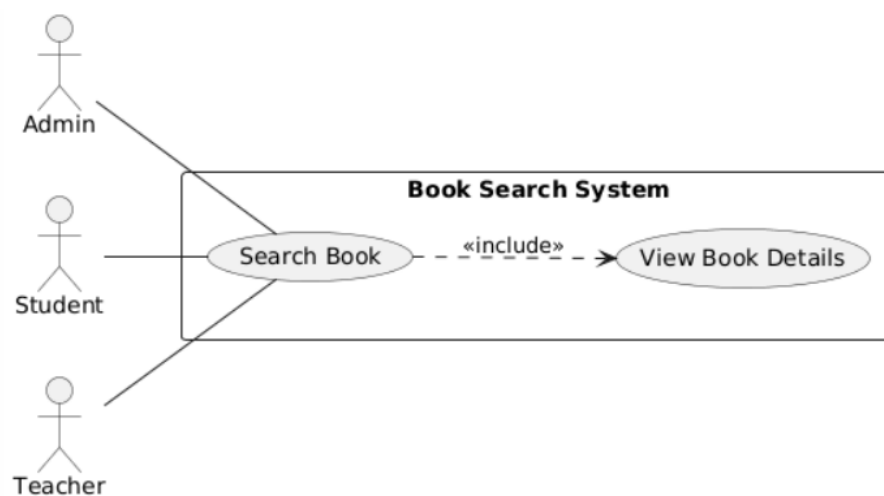


Fig: Book searching use case diagram

5.1.2.5 Report and Monitoring Subsystem

Description

This subsystem supports administrative reporting and monitoring of library usage and transactions.

Tasks

- Generate Issued Book Report
- Generate Returned Book Report
- View Outside Book Logs

Actor: Admin

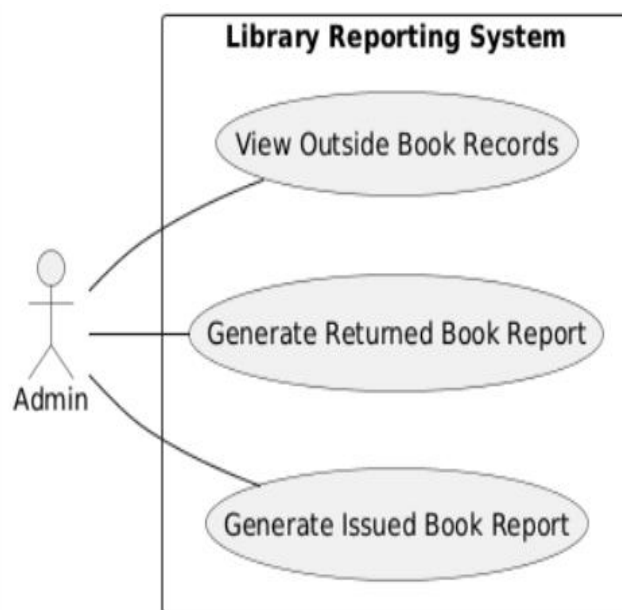


Fig: Report generate use case diagram

5.2 Data Flow Diagrams:

A Data Flow Diagram (DFD) is a graphical representation of how data flows within a system. It illustrates the movement of data between external entities, internal processes, and data stores, allowing stakeholders to understand system operations without focusing on implementation details.

For the **IICT Library Management System**, the DFD explains how member information, book records, borrowing transactions, outside book logs, and overdue

reports are processed and maintained. It shows how students, faculty members, library staff, and administrators interact with the system to perform daily library activities efficiently.

The DFD is divided into different levels:

- **Level 0:** Presents an overall view of the system and its external interactions.
 - **Level 1:** Breaks down the system into major functional modules.
 - **Level 2:** Further decomposes major modules into detailed subprocesses.
-

Description

The IICT Library Management System DFD focuses on the following major subsystems:

1. **External Entities:** Students/Faculty, Library Staff, and IICT Administration.
2. **Processes:** Member management, inventory and catalog management, circulation management, outside book tracking, and overdue reporting.
3. **Data Stores:** Member database, book database, borrowing records, and outside book logs.

The DFD ensures proper data handling, minimizes record duplication, improves circulation efficiency, and provides accurate reporting for administrative decision-making.

5.1.1 Level 0 Data Flow Diagram (DFD)

System Overview

The Level 0 DFD represents the IICT Library Management System as a single process interacting with external entities.

External Interactions:

- Students and faculty submit borrowing/return requests and search queries to the system.
- Library staff manage circulation and inventory updates.
- The IICT Administration receives analytical reports and provides system policies.

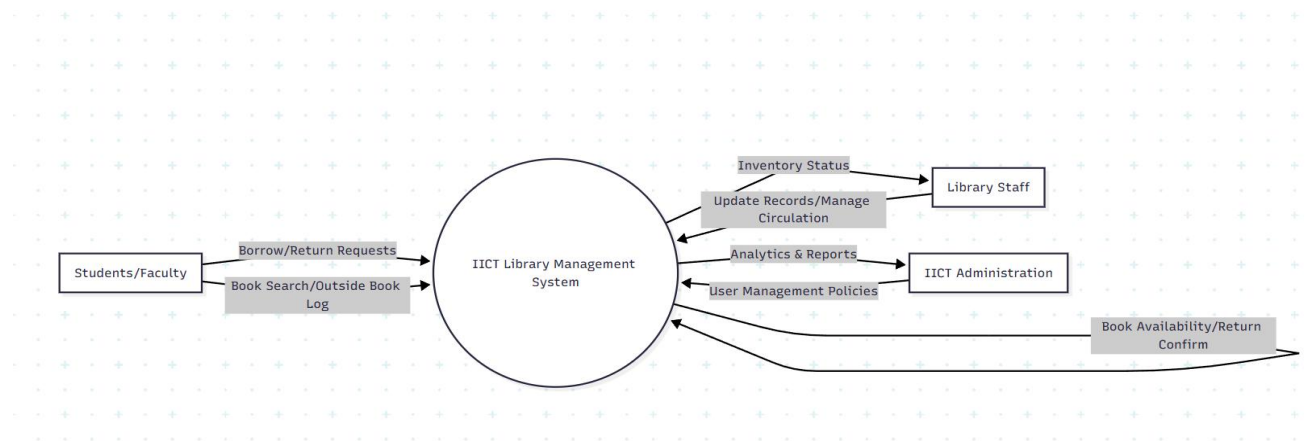


Figure: Data Flow Diagram Level 0 – System Overview

This diagram presents only the major inputs and outputs without showing internal processing details.

5.2.2 Level 1 Data Flow Diagram (DFD)

Major Subsystem Decomposition

The Level 1 DFD decomposes the main system into five core processes:

Process	Description
1.0 Member Management	Registers and maintains member records of students and faculty.
2.0 Inventory & Catalog Management	Maintains book information and search facilities.
3.0 Circulation Management	Handles book borrowing and returning transactions.
4.0 Outside Book Tracking	Logs books brought from outside and monitors their exit.
5.0 Overdue & Reporting	Generates overdue alerts and analytical reports.

Key Data Stores:

- Member Database
- Book Database
- Borrowing Records
- Outside Book Log

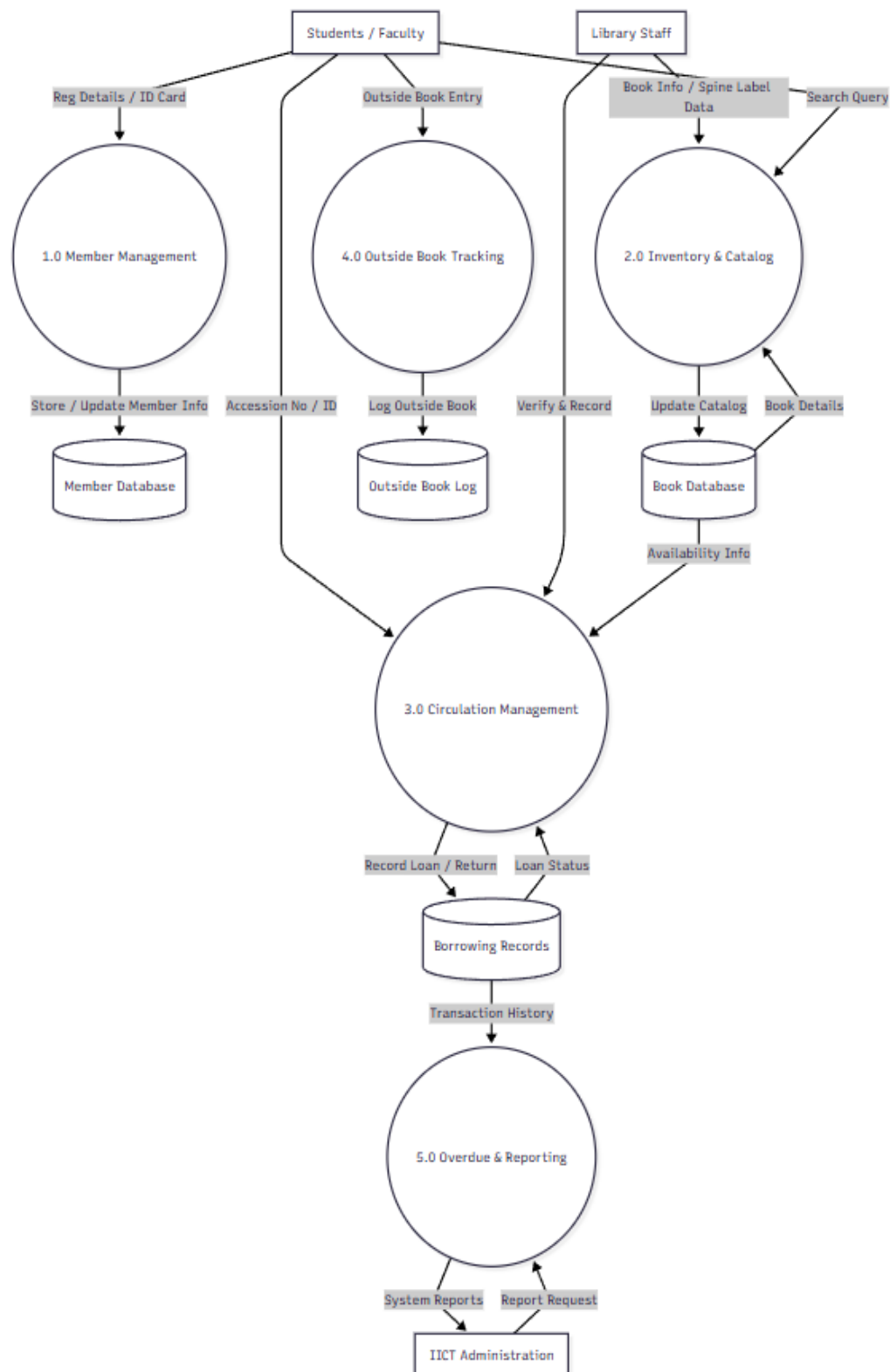


Figure: *Data Flow Diagram Level 1 – System Overview*

This level provides a clear picture of how all major library operations interact with data stores and external entities.

5.2.3 Level 2 Data Flow Diagram (DFD)

Detailed Subprocess View

The Level 2 DFD further decomposes critical Level 1 processes into detailed subprocesses:

A. Circulation Management (Process 3.0)

Subprocess	Description
3.1 Verify Identity	Confirms member validity using ID and member database.
3.2 Record Borrowing	Records borrowing information and updates book availability.
3.3 Process Return	Updates return status and clears borrowed entries.
3.4 Update Due Status	Identifies overdue loans and notifies staff.

Data flows between the **Member Database**, **Book Database**, and **Borrowing Records** to ensure transaction accuracy.

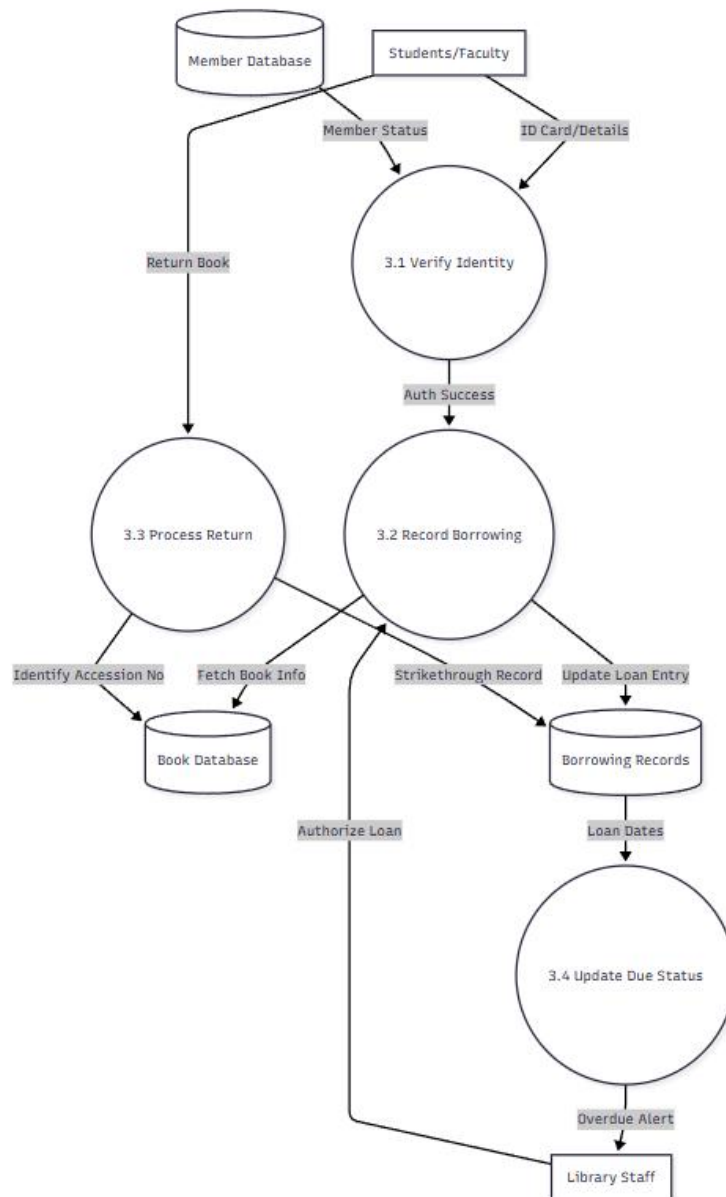


Figure: Data Flow Diagram Level 2 – Circulation Management

B. Outside Book Tracking (Process 4.0)

Subprocess

Description

4.1 Log Entry

Records details of outside books brought by students.

Subprocess	Description
4.2 Verify Outside Book	Displays active logs for monitoring purposes.
4.3 Log Exit	Updates log when books leave the library premises.

All records are maintained in the **Outside Book Log** data store.

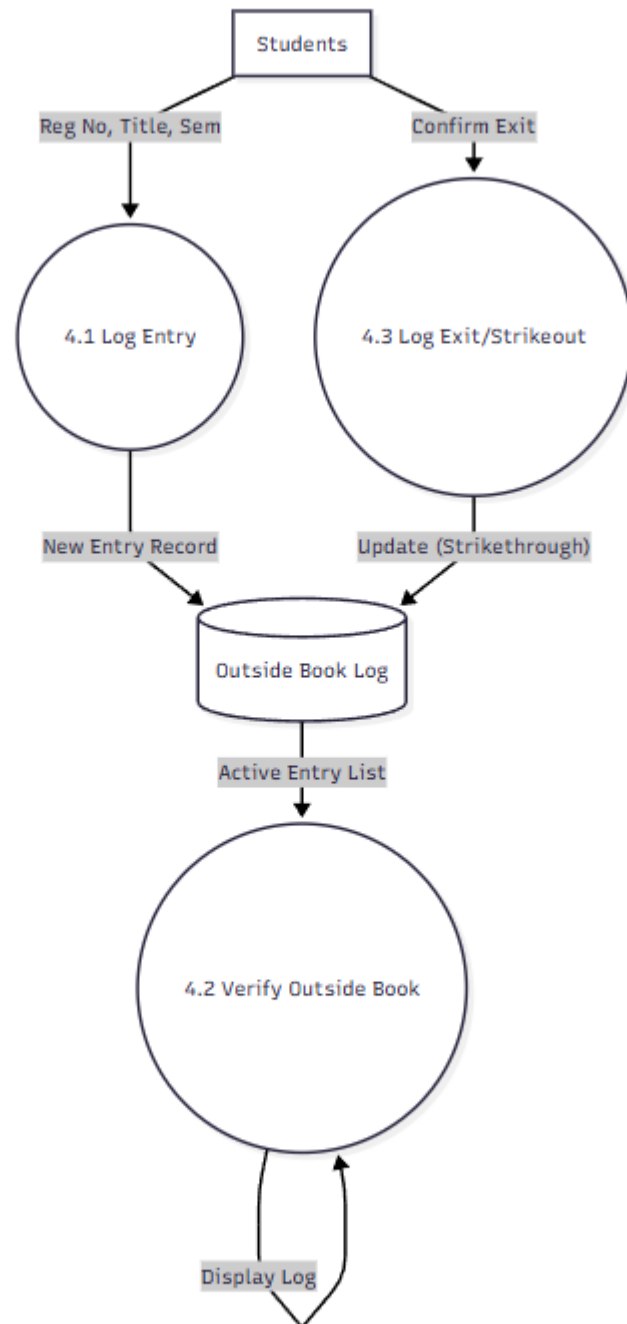


Figure: *Data Flow Diagram Level 2 – Outside Book Tracking*

This level shows how individual actions are processed at the lowest logical level, ensuring accountability, security, and efficient library management.

5.3 Class Diagrams:

5.3.1 Authentication & User Role Management

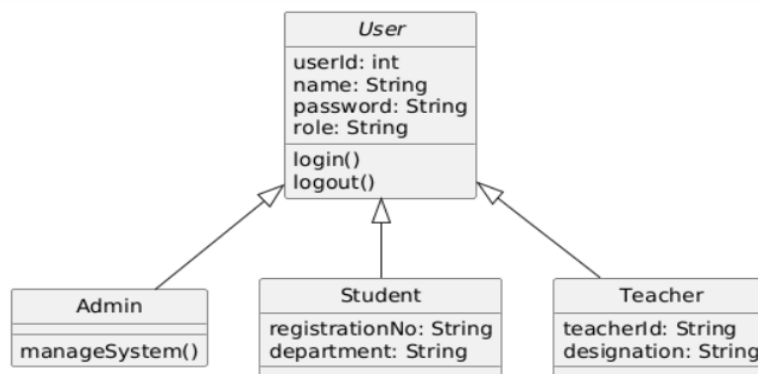
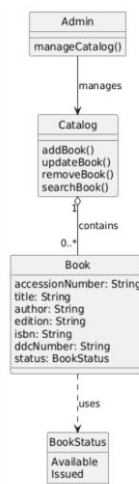


Figure: Authentication and user role management diagram

5.3.2 Book Catalog & Classification Process



5.3.3 Borrowing & Returning Process

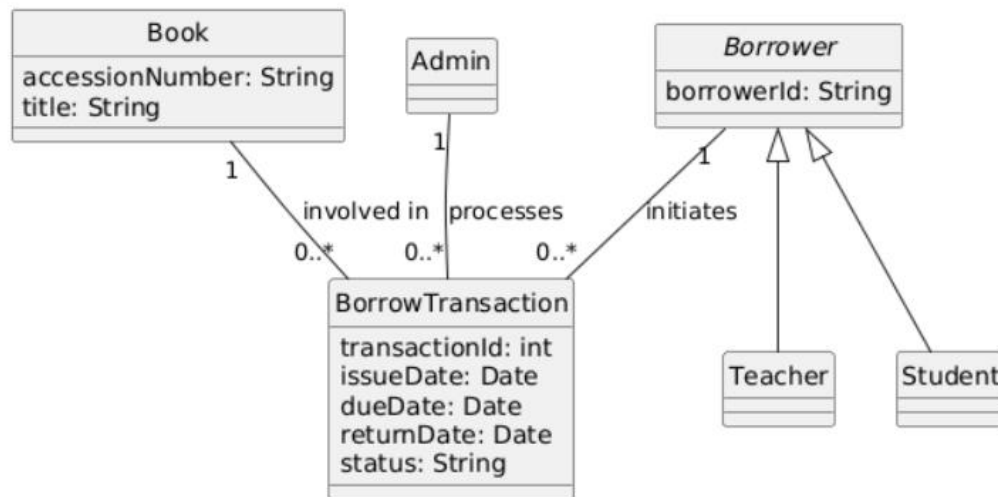


Figure: Borrowing and Returning process

5.3.4 Outside Book Monitoring Process

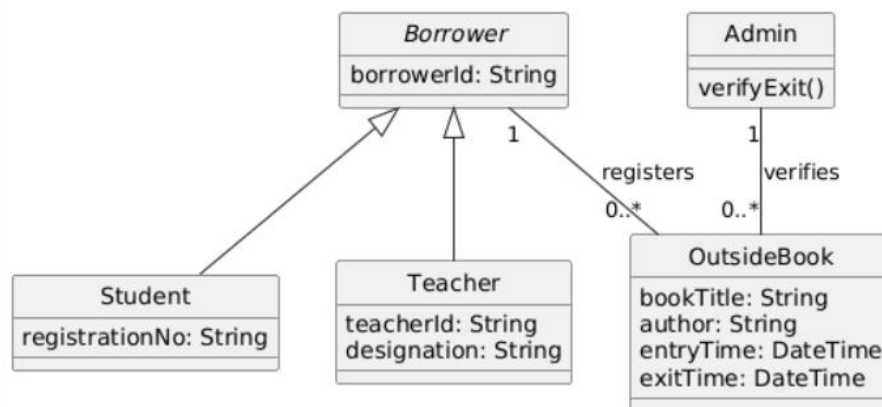


Fig: Outside book monitoring process

5.3.5 UML Class diagram for Level 1 System

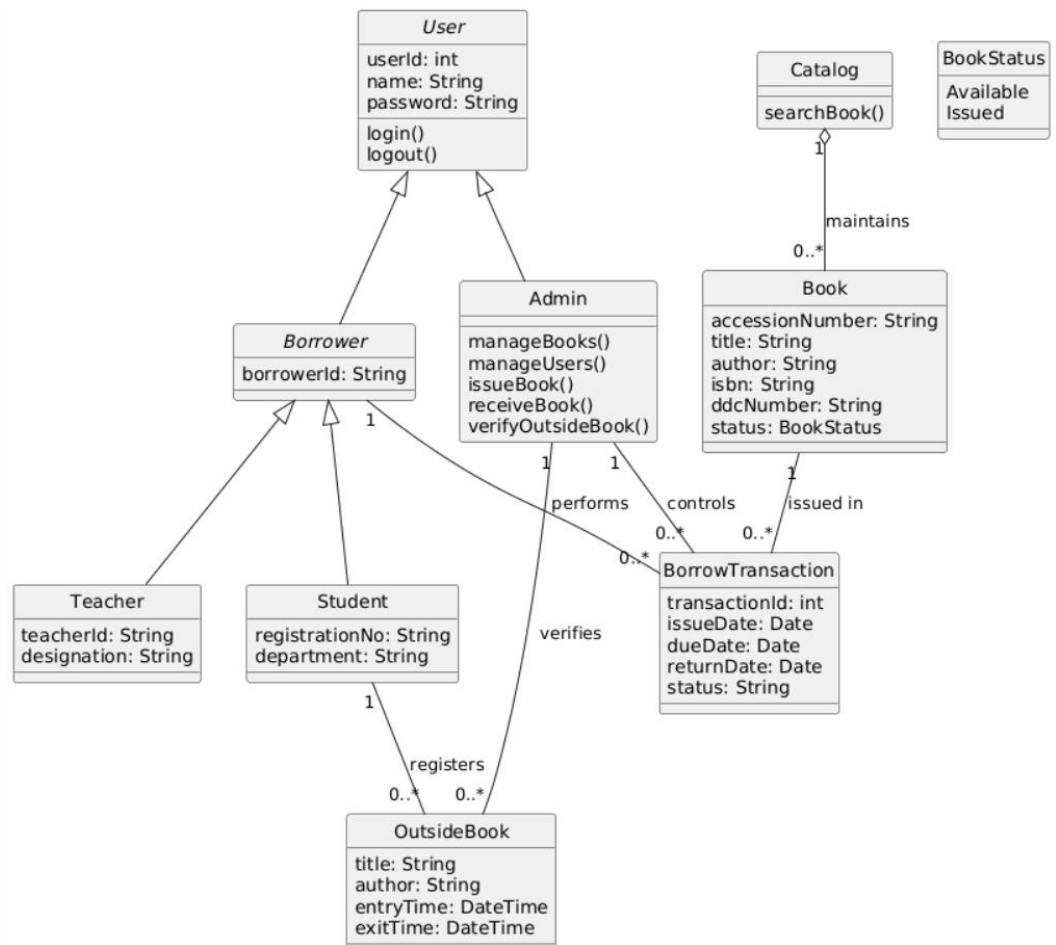


Fig: Full system class diagram

6.Conclusion

The Software Requirements Specification (SRS) for the **IICT Library Management System** provides a complete and structured description of the functional and non-functional requirements needed to develop a reliable and efficient digital library system for IICT. This document clearly defines system objectives, user roles, operational workflows, security constraints, and performance expectations, ensuring that all stakeholders share a common understanding of the system's purpose and behavior.

The proposed system addresses the limitations of traditional manual library operations by introducing a centralized, web-based platform that improves accuracy, transparency, and efficiency in book cataloging, circulation management, borrower tracking, and outside book monitoring. By incorporating role-based access control, Dewey Decimal Classification, secure authentication, and detailed reporting mechanisms, the system preserves real-world library practices while enhancing data reliability and accountability.

This SRS serves as a formal baseline for system design, implementation, testing, and evaluation. It will guide developers throughout the development life cycle and support future scalability and maintenance. Ultimately, the IICT Library Management System is expected to significantly modernize library operations, reduce administrative workload, and provide students and faculty with faster, safer, and more convenient access to academic resources.